Rules of Thumb

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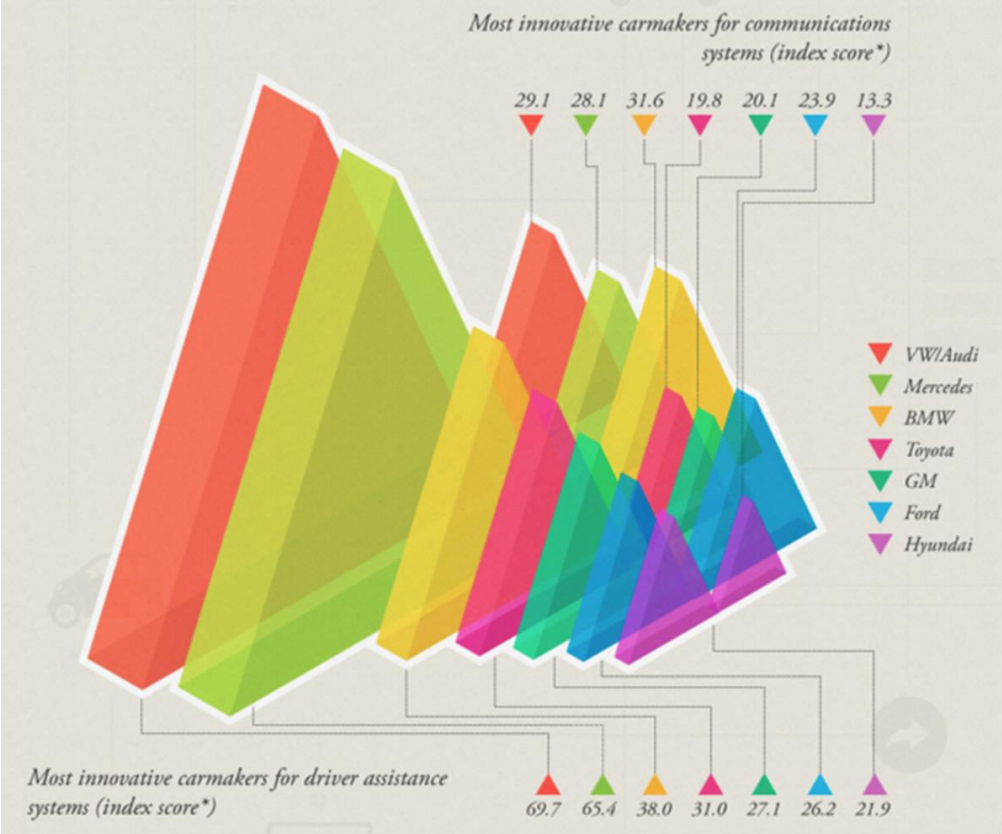
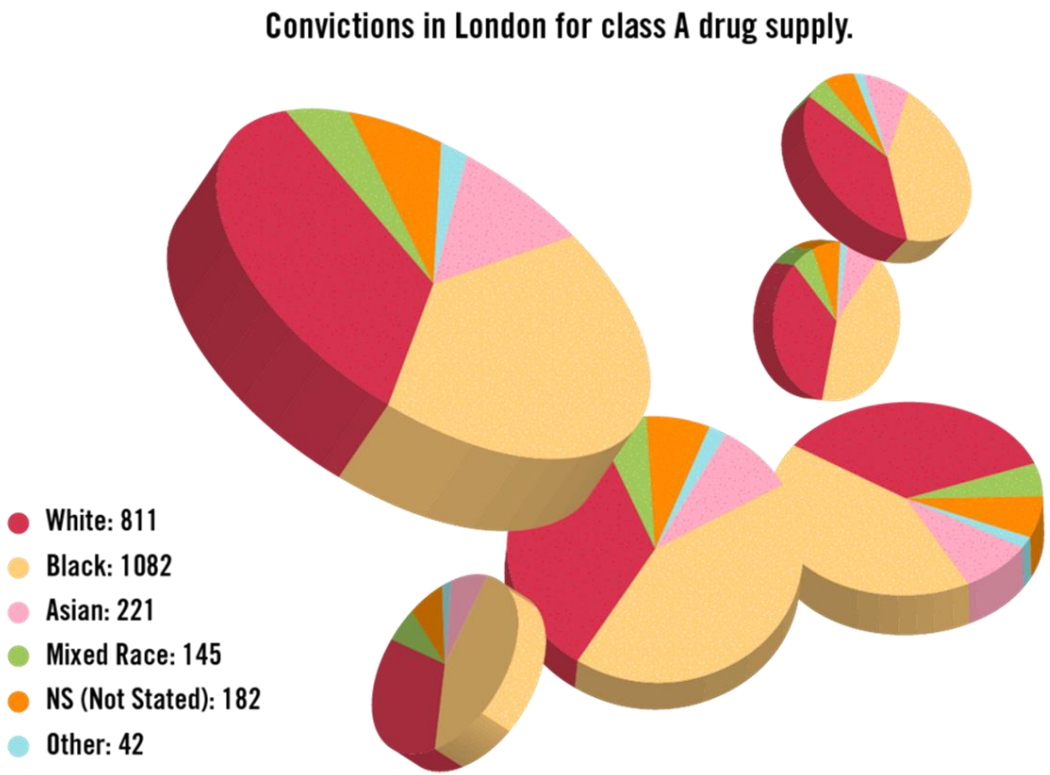
Slides taken, reformatted, and used from Tamara Munzner (UBC, Canada)

Rules of thumb

- Guidelines and considerations, not absolute rules
  - When to use 3D? When to use 2D?
  - When to use eyes instead of memory?
  - When does immersion help?
  - When to use overviews?
  - How long is too long?
  - Which comes first, form or function?

Unjustified 3D is all too common in the news and elsewhere



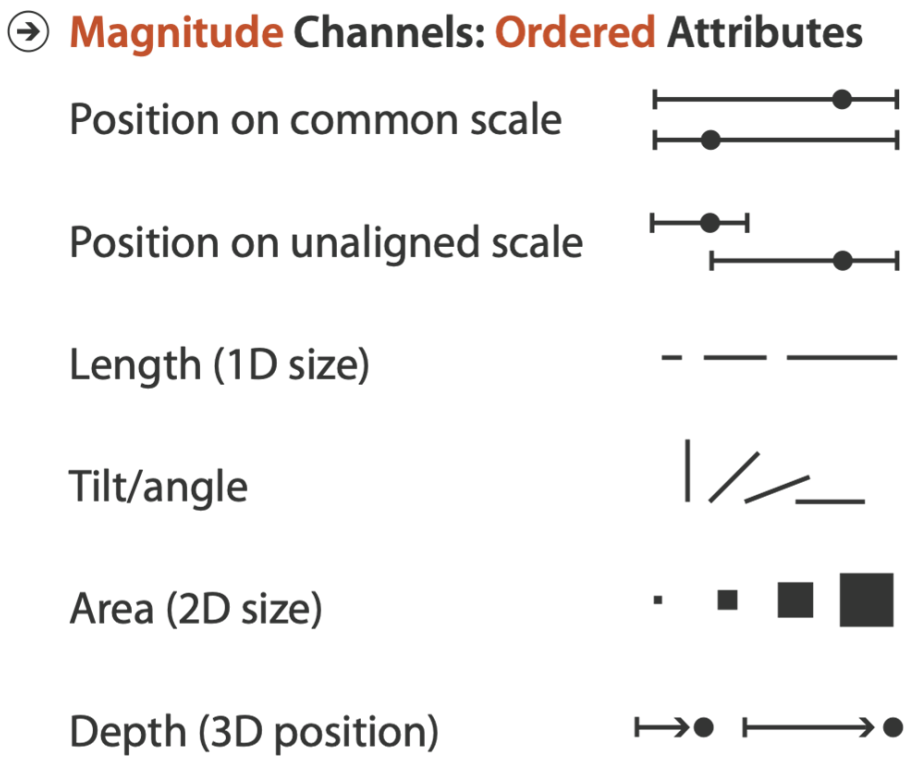


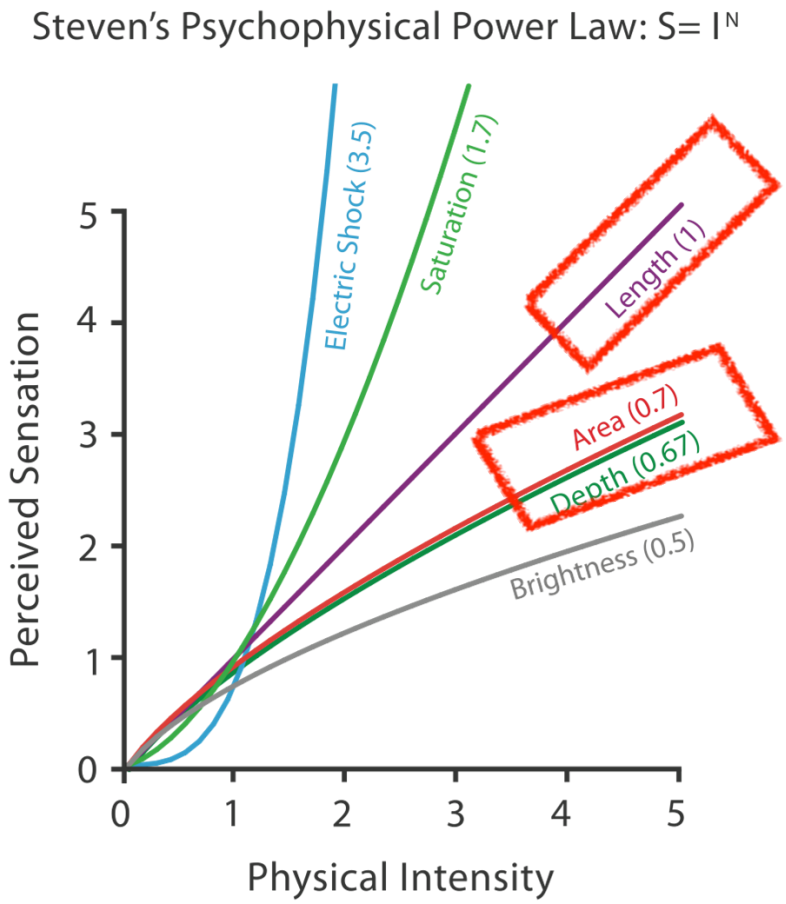
http://viz.wtf/post/137826497077/eye-popping-3d-triangles

http://viz.wtf/post/139002022202/designer-drugs-ht-ducqn

Depth vs power of the plane

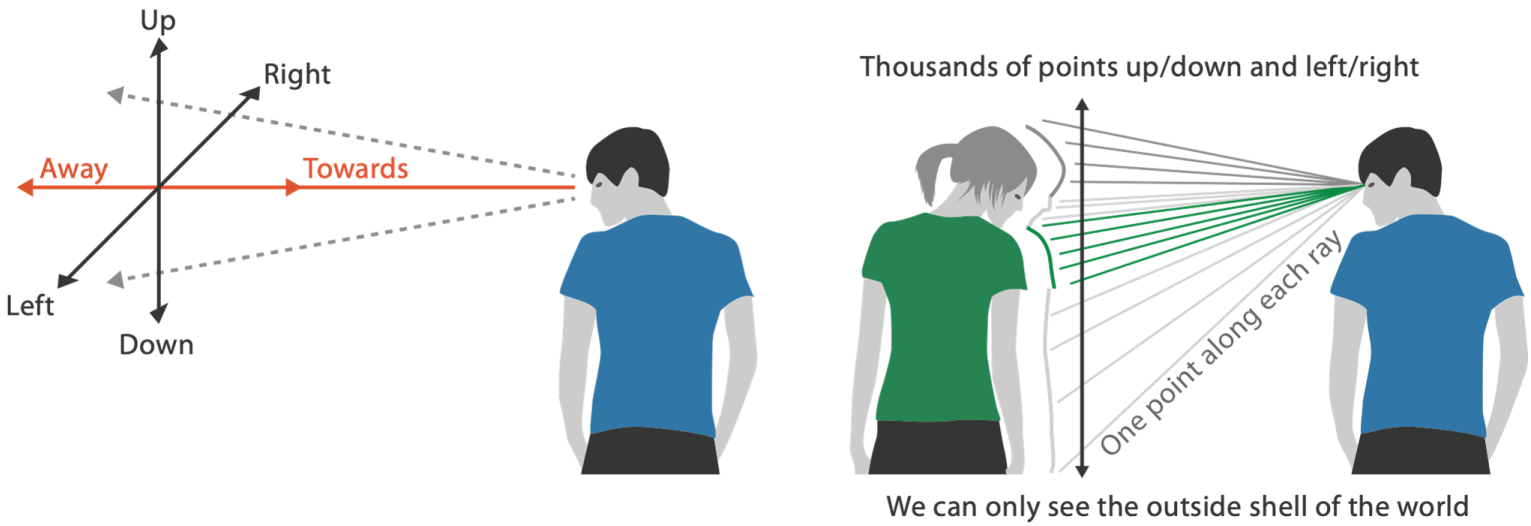
- High-ranked spatial position channels: planar spatial position – not depth





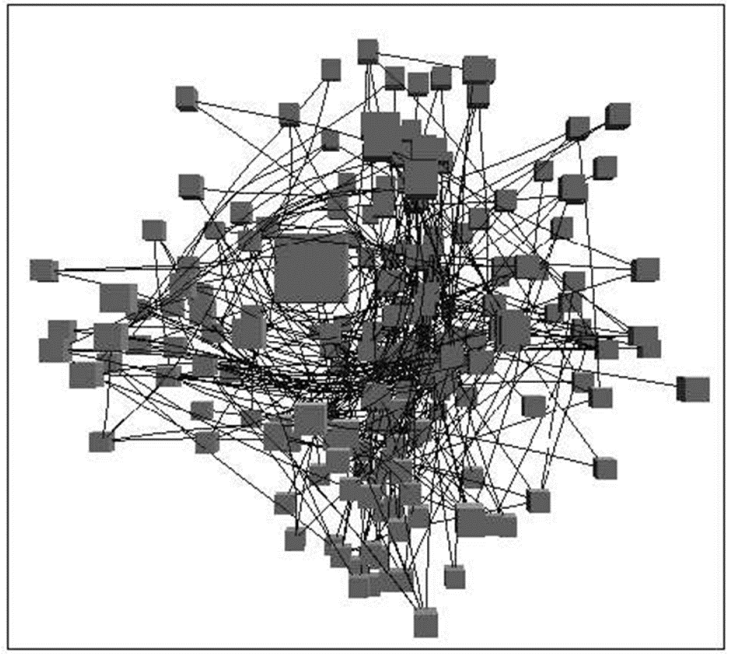
No unjustified 3D: danger of depth

- We do not really live in 3D: we see in 2.05 D
  - Acquire more information on the image plane quickly from eye movements
  - Acquire more information for depth slower, from head/body motion



Occlusion hides information

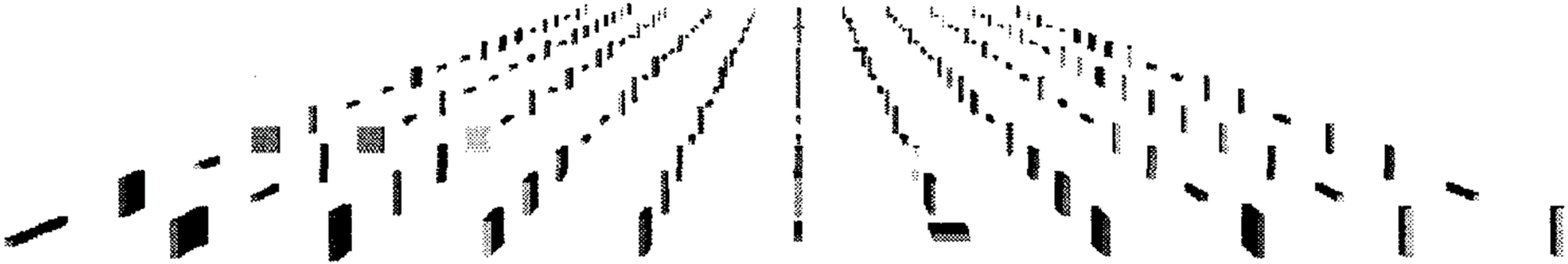
- Occlusion
- Interaction can resolve, but at the cost of time and cognitive load



[Distortion Viewing Techniques for 3D Data. Carpendale et al. InfoVis1996.]

Perspective distortion loses information

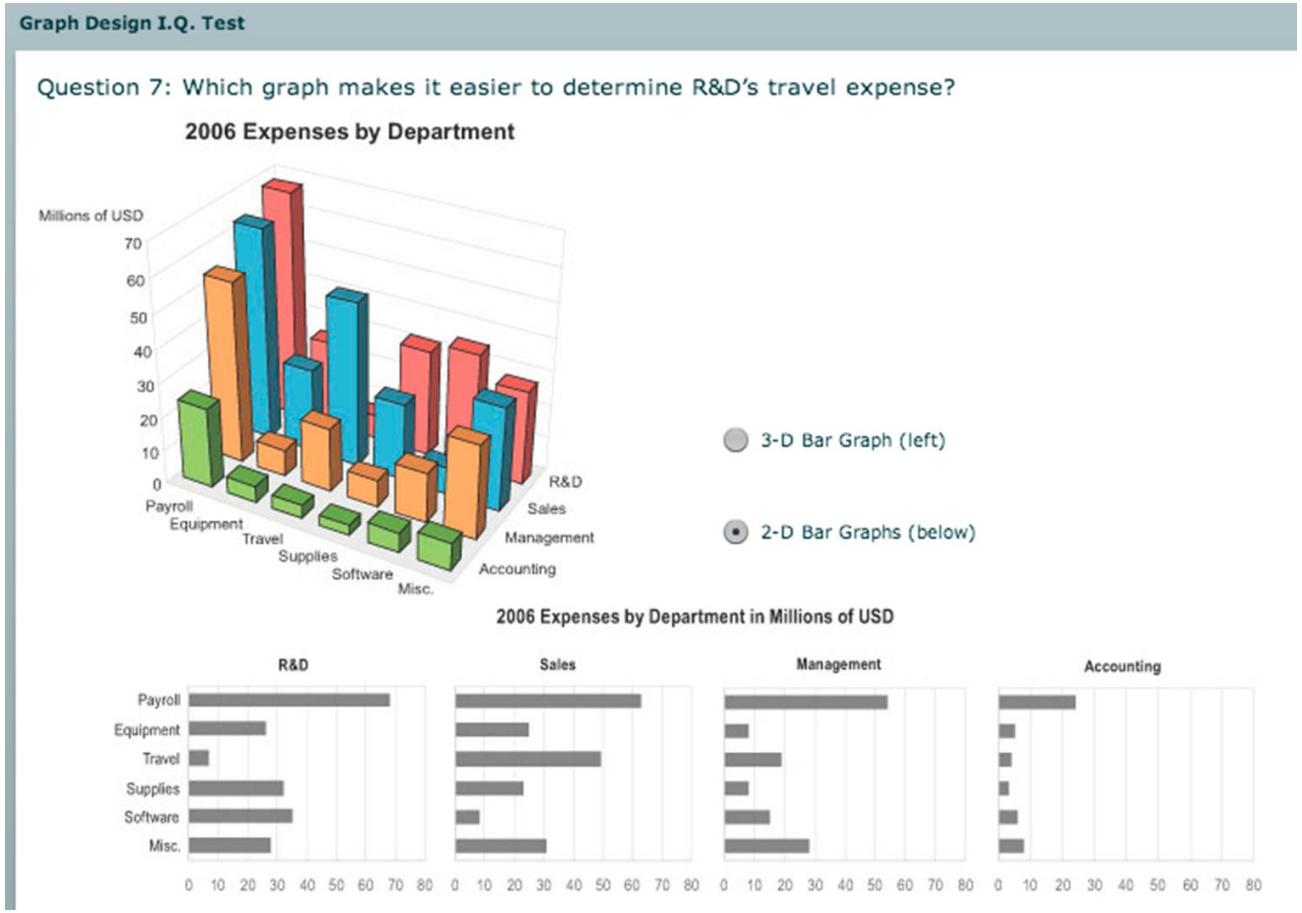
- Perspective distortion
  - Interferes with all size channel encodings
  - Power of the plane is lost!



[Visualizing the Results of Multimedia Web Search Engines. Mukherjea, Hirata, and Hara. InfoVis 96]

3D vs 2D bar charts

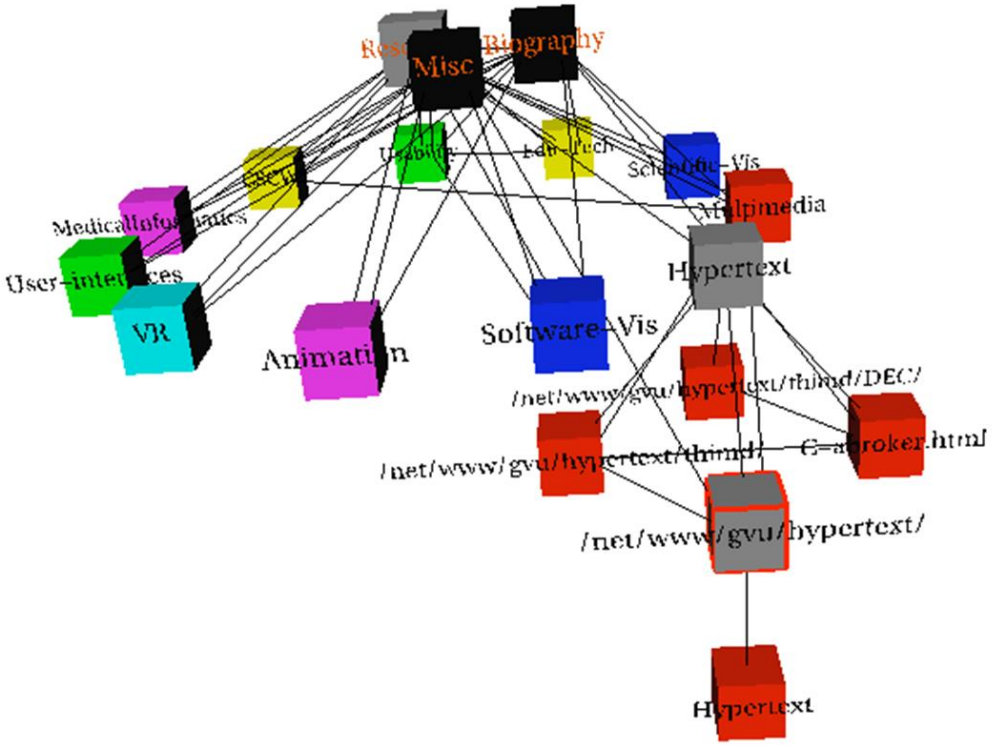
- 3D bars: very difficult to justify!
  - Perspective distortion
  - Occlusion
- Faceting into 2D almost always better choice



[http://perceptualedge.com/files/GraphDesignIQ.html]

Tilted text is not legible

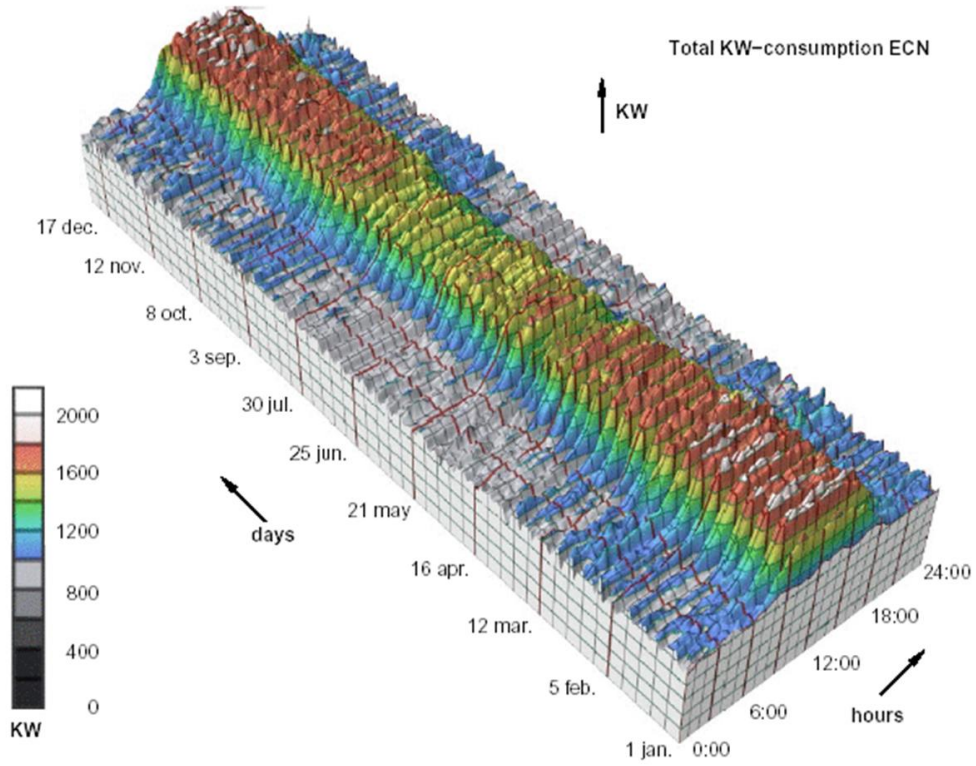
- Text legibility
  - Far worse when tilted from image plane



[*Visualizing the World-Wide Web with the Navigational View Builder. Mukherjea and Foley. Computer Networks and ISDN Systems, 1995.*]

No unjustified 3D example: time

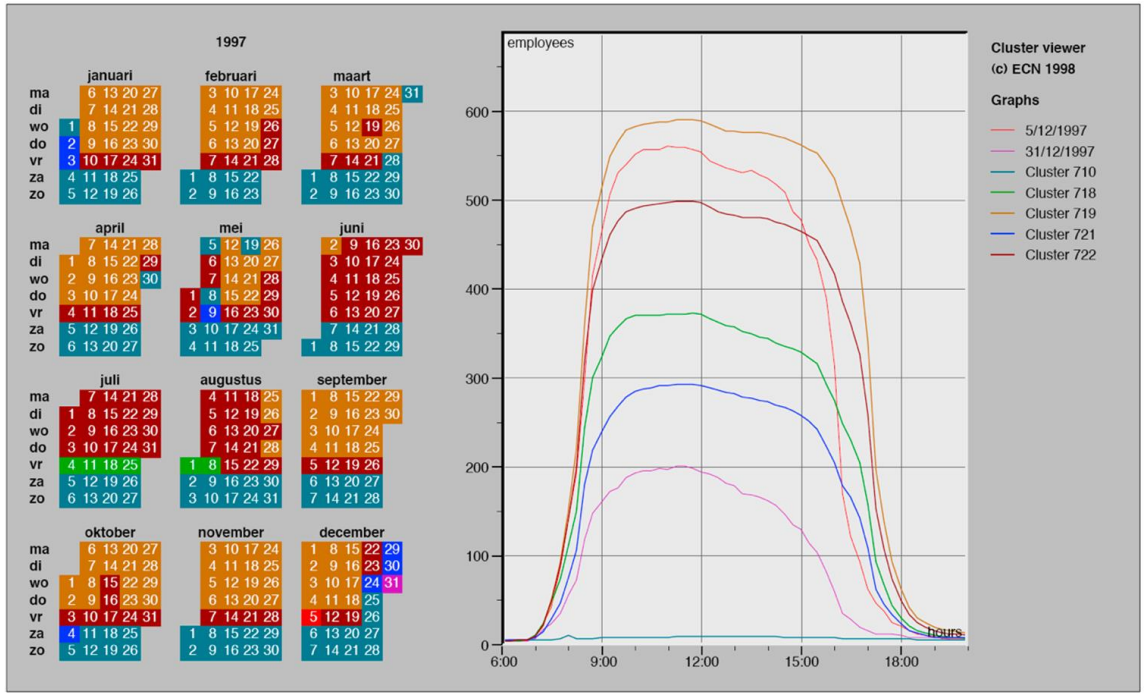
- Extruded curves: detailed comparisons impossible



[Cluster and Calendar based Visualization of Time Series Data. van Wijk and van Selow, Proc. InfoVis 99.]

No unjustified 3D example: transform for new data abstraction

- Derived data: cluster hierarchy
- Juxtapose multiple views: calendar, super imposed 2D curves



[Cluster and Calendar based Visualization of Time Series Data. van Wijk and van Selow, Proc. InfoVis 99.]

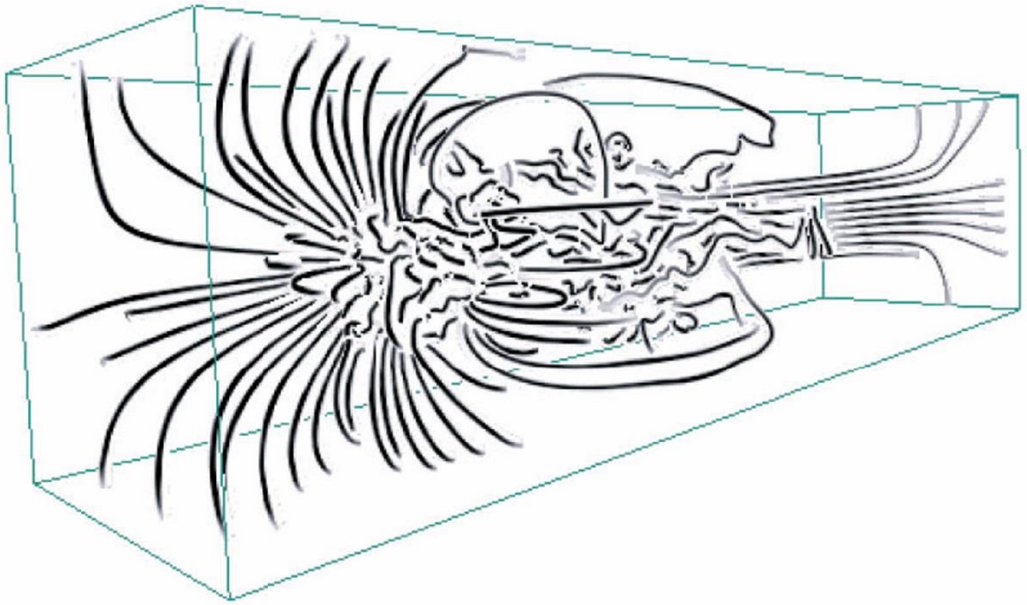
Justified 3D: shape perception

Targets

Spatial Data

Shape

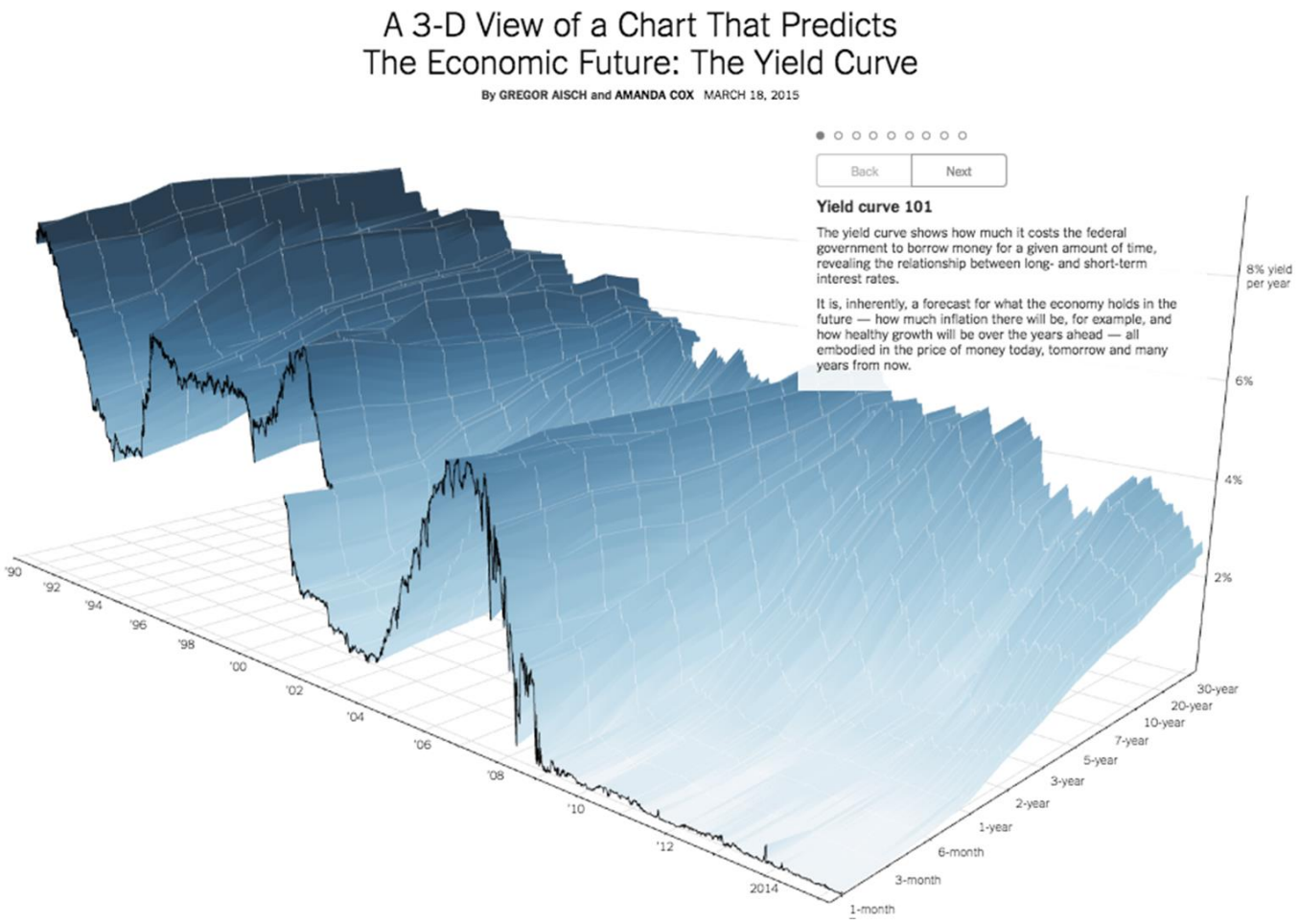
- Benefits outweigh costs when the task is shape perception for 3D spatial data
  - Interactive navigation supports synthesis across many viewpoints



[Image-Based Streamline Generation and Rendering. Li and Shen. IEEE Trans. Visualization and Computer Graphics (TVCG) 13:3 (2007), 630–640.]

Justified 3D: Economic growth curve

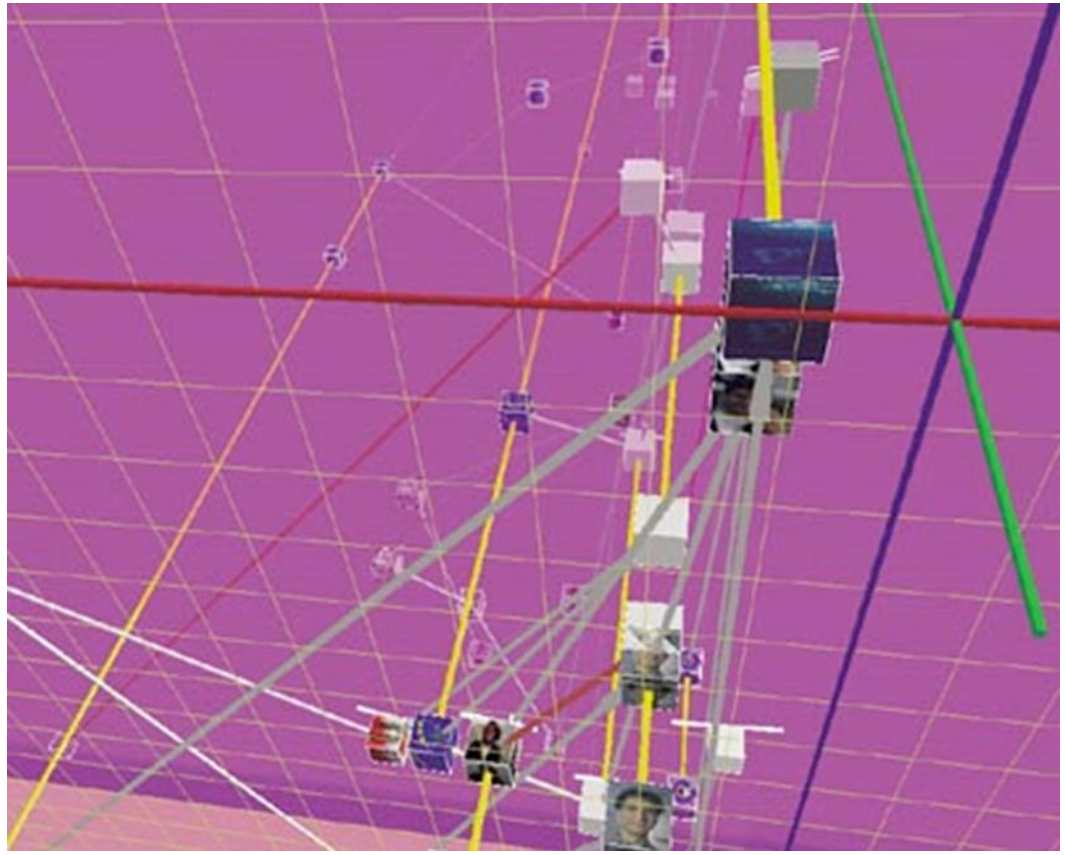
- Constrained navigation steps through carefully designed view points



http://www.nytimes.com/interactive/2015/03/19/upshot/3d-yield-curve-economic-growth.html

No unjustified 3D

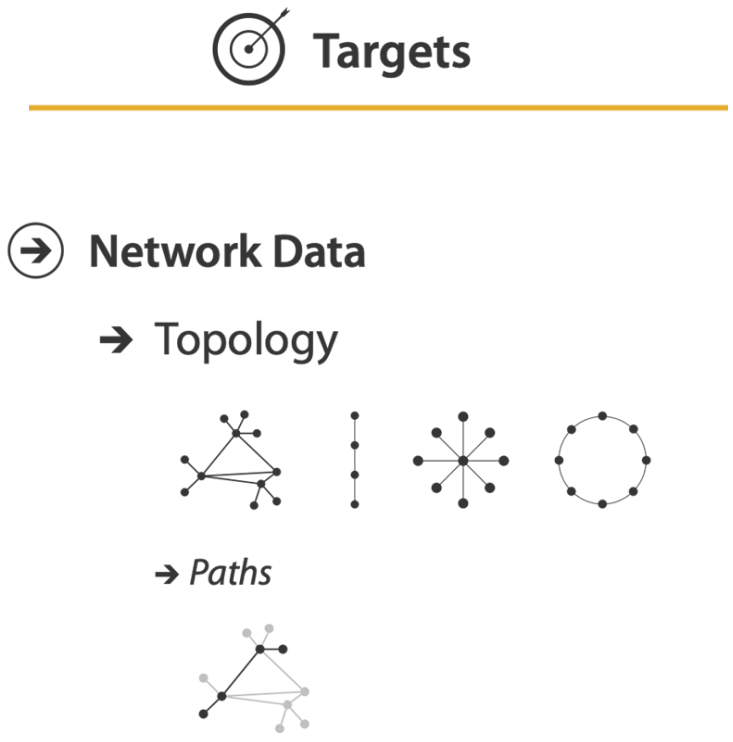
- 3D legitimate for true 3D spatial data
- 3D needs very careful justification for abstract data
  - Enthusiasm in 1990s, but now skepticism
  - Be especially careful with 3D for point clouds or networks



[WEBPATH-a three dimensional Web history. Frecon and Smith. Proc. InfoVis 1999]

No unjustified 2D

- Consider whether network data requires a 2D spatial layout
  - Especially if the reading text is central to the task!
  - Arranging as a network means lower information density and harder label lookup compared to text lists
- Benefits outweigh costs when topological structure/context important for the task
  - Be especially careful with search results, document collections, ontologies



Eyes beat memory

- Principle: external cognition vs internal memory
  - Easy to compare by moving eyes between side-by-side views
  - Harder to compare visible item to memory of what you saw
- Implications for animation
  - Great for choreographed story telling
  - Great for transitions between two states
  - Poor for many states with changes everywhere
    - Consider small multiples instead

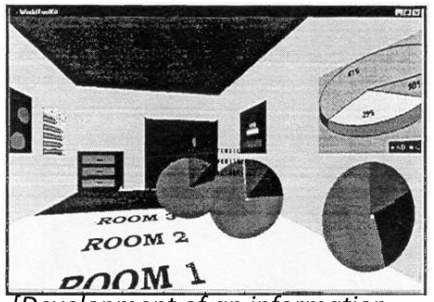
literal | abstract

animation | small multiples

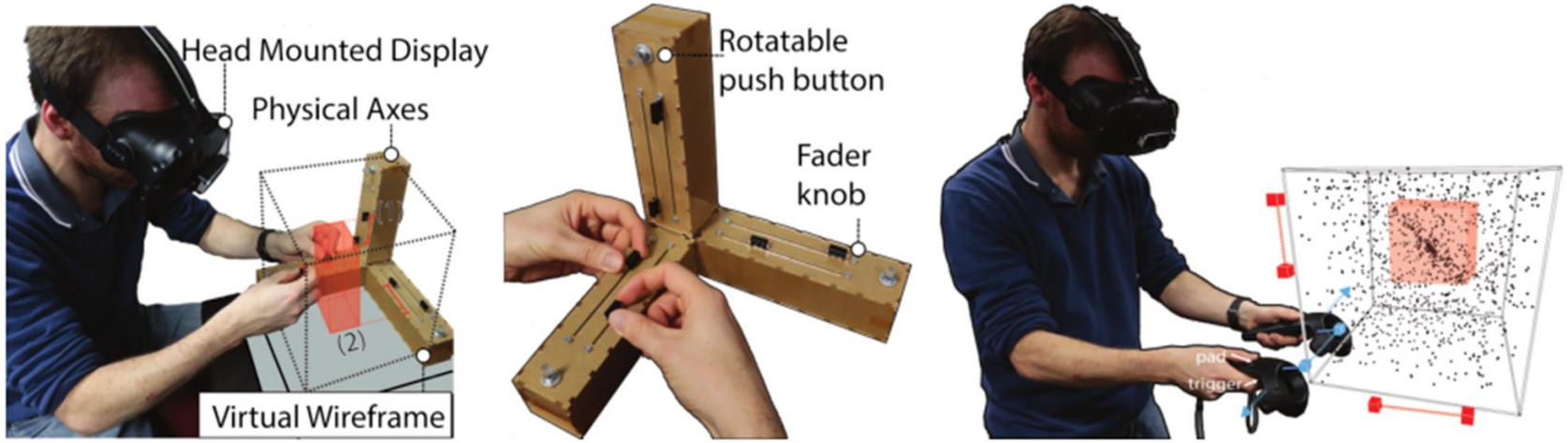
show time with time ⟷ show time with space

Resolution beats immersion

- Immersion typically not helpful for abstract data
  - Do not need a sense of presence or stereoscopic 3D
  - Desktop is also better for workflow integration
- Resolution much more important: pixels are the scarcest resource
- First wave: virtual reality for abstract data difficult to justify
- Second wave: AR/MR (augmented/mixed reality) has more promise



[Development of an information visualization tool using virtual reality. Kirner and Martins. Proc. Symp. Applied Computing 2000]

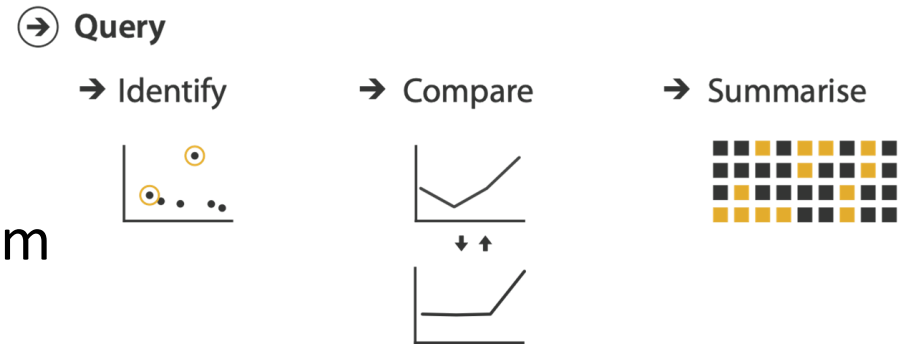


[A Design Space for Spatio-Data Coordination: Tangible Interaction Devices for Immersive Information Visualisation. Cordeil, Bach, Li, Elliott, and Dwyer. Proc. PacificVis 2017 Notes.]

Overview first, zoom and filter, details on demand

- Influential mantra from Schneiderman

[The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations. Shneiderman. Proc. IEEE Visual Languages, pp. 336–343, 1996.]



- Overview = summary
  - Microcosm of full visualisation design problem

Rule of thumb: Responsiveness is required

- Visual feedback: three rough categories
  - 0.1 seconds: perceptual processing
    - Subsecond response for mouseover highlighting – ballistic motion
  - 1 second: immediate response
    - Fast response after a mouse click, button press
  - 10 seconds: brief tasks
    - Bounded response after dialog box – a mental model of heavy-weight operation (file load)
- Scalability considerations
  - Highlight selection without complete redraw of view (graphics front buffer)
  - Show hourglass for multi-second operations (check for cancel/undo)
  - Show progress bar for long operations (the process in a background thread)
  - Rendering speed when item count is large (guaranteed frame rate)

Function first, form next

- Dangerous to start with aesthetics
  - Usually impossible to add a function retroactively
- Start with a focus on functionality
  - Possible to improve aesthetics later on, as refinement
  - If no expertise in-house, find good graphic designer to work with
  - Aesthetics do matter! Another level of function
    - Visual hierarchy, alignment, flow

Best practices: labelling

- Make visualizations as self-documenting as possible
  - Meaningful and useful title, labels, legends
    - Axes and panes/subwindows must have labels
      - And axes should have good min/max boundary tick marks
    - Everything that’s plotted should have a legend
      - And own header/labels if not redundant with main title
    - Use a reasonable numerical format
      - Avoid scientific notation in most cases

Rules of thumb

- No unjustified 3D
  - Power of the plane
  - Disparity of depth
  - Occlusion
  - Perspectives distortion dangers
  - Tilted text is not legible
- No unjustified 2D
- Eyes beat memory
- Resolution over immersion
- Overview first, zoom and filter, details on demand
- Responsiveness is required
- Function first, form next